

MARTIN CARRASCO

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Education

2029 (Expected)	Ph.D. in Computer Science (ELLIS)	University of Fribourg
2025	M.S. in Artificial Intelligence	VU Amsterdam
2023	B.S. in Computer Science	Universidad de Ingeniería y Tecnología

Research Interests

Geometric/Topological Deep Learning; Representation Learning; Graph Machine Learning

Publications

The symbol * denotes *equal* first-author contribution while the symbol † denotes *equal* supervision contribution.

Conference & Journal

- 2025 [1] L. Telyatnikov, G. Bernardez, M. Montagna, M. Hajj, **M. Carrasco**, P. Vasylenko, M. Papillon, G. Zamzmi, M. T. Schaub, J. Verhellen, P. Snopov, B. Miquel-Oliver, M. Gil-Sorribes, A. Molina, V. Guallar, T. Long, J. Suk, P. Rygiel, A. V. Nikitin, G. Escalona, M. Banf, D. Filipiak, L. Imasheva, M. Schattauer, A. L. Martinez, H. Fritze, M. Masden, V. Sa nchez, M. Lecha, A. Cavallo, C. Battiloro, M. Piekenbrock, M. Tec, G. Dasoulas, N. Miolane, S. Scardapane, and T. Papamarkou, “[Topobench: A framework for benchmarking topological deep learning](#),” *Journal of Data-centric Machine Learning Research*, 2025.
- 2024 [2] M. G. Palma, J. Gonzalez, **M. Carrasco**, R. Rubio-Noriega, K. Bergman, and R. Azevedo, “[Inter-node Message Passing through Optical R econfigurable Memory Channel](#),” *IEEE Access*, pp. 1–1, 2024.

Workshops

- 2026 [3] K. Limbeck, N. Häusermann, **M. Carrasco**, G. Wolf[†], and B. Rieck[†], “Diversity curves for graph representation learning,” in *ECML PKDD Workshop on Mining and Learning with Graphs*, 2026. arXiv: 2605.06466 [cs.LG].
- 2026 [4] J. S. Schmidt^{*}, **M. Carrasco**^{*}, E. Röell, G. Wolf, N. Blaser[†], and B. Rieck[†], “Topology and combinatorics: Generalization in deep learning,” in *Symposium on Geometry Processing Posters*, M. Skouras and M. Livesu, Eds., 2026.
- 2025 [5] **M. Carrasco**^{*}, V. A. Martirosyan^{*}, A. Mehrab^{*}, C. Netto^{*}, E. Okoyomon^{*}, and C. Graziani, “[Rademacher Meets Colors: More Expressivity, but at W hat Cost?](#)” *New Perspectives in Graph Machine Learning*, 2025.

- 2024 [6] G. Bernárdez, L. Telyatnikov, M. Montagna, F. Baccini, M. Papillon, M. Ferriol-Galmés, M. Hajij, T. Papamarkou, M. S. Bucarelli, O. Zaghen, J. Mathe, A. Myers, S. Mahan, H. Lillemark, S. Vadgama, E. Bekkers, T. Doster, T. Emerson, H. Kvinge, K. Agate, N. K. Ahmed, P. Bai, M. Banf, C. Battiloro, M. Beketov, P. Bogdan, **M. Carrasco**, A. Cavallo, Y. Y. Choi, G. Dasoulas, M. Elphick, G. Escalona, D. Filipiak, H. Fritze, T. Gebhart, M. Gil-Sorribes, S. Goomanee, V. Guallar, L. Imasheva, A. Irimia, H. Jin, G. Johnson, N. Kanakaris, B. Koloski, V. Kovac, M. Lecha, M. Lee, P. Leroy, T. Long, G. Magai, A. Martinez, M. Masden, S. Meznar, B. Miquel-Oliver, A. Molina, A. Nikitin, M. Nurisso, M. Piekenbrock, Y. Qin, P. Rygiel, A. Salatiello, M. Schattauer, P. Snopov, J. Suk, V. Sánchez, M. Tec, F. Vaccarino, J. Verhellen, F. Wantiez, A. Weers, P. Zajec, B. Skrlj, and N. Miolane, “[ICML Topological Deep Learning Challenge 2024: Beyond the Graph Domain](#),” *Proceedings of Machine Learning Research*, vol. 251, S. Vadgama, E. Bekkers, A. Pouplin, S.-O. Kaba, R. Walters, H. Lawrence, T. Emerson, H. Kvinge, J. Tomczak, and S. Jegelka, Eds., pp. 420–428, Jul. 2024.
- 2024 [7] **M. Carrasco**, A. Berentzen, and A. Garcia, *Applications of TopoX to Equivariant Topological Networks*, Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM) @ ICML, Jul. 2024.

Preprints

- 2026 [8] **M. Carrasco**, V. A. Martirosyan, A. Mehrab^{*}, C. Netto^{*}, E. Okoyomon^{*}, and C. Graziani, “On the Rademacher Complexity of GNNs: Unifying Expressivity and Geometry,” *Preprint*, 2026.
- 2026 [9] **M. Carrasco**^{*}, O. Zaghen^{*}, K. Sumaraj, E. Bekkers, and B. Rieck, “[Graph Homomorphism Distortion: A Metric to Distinguish Them All and in the Latent Space Bind Them](#),” *arXiv preprint arXiv:2511.03068*, 2026.
- 2026 [10] K. Limbeck, N. Häusermann, **M. Carrasco**, G. Wolf, and B. Rieck, “[Diversity Curves for Graph Representation Learning](#),” *arXiv preprint arXiv:2605.06466*, 2026.
- 2026 [11] J. S. Schmidt^{*}, **M. Carrasco**^{*}, E. Röell, G. Wolf, N. Blaser[†], and B. Rieck[†], “[No Triangulation Without Representation: Generalization in Topological Deep Learning](#),” *arXiv preprint arXiv:2605.06467*, 2026.
- 2025 [12] G. Bernardez, M. Montagna, L. V. Langendonck, **M. Carrasco**, A. Akbari, L. Cornelis, M. Papillon, P. Barlet-Ros, N. Miolane, and L. Telyatnikov, “[HOPSE: Scalable Higher-Order Positional and Structural Encoder for Combinatorial Representations](#),” *arXiv preprint arXiv:2505.15405*, 2025.

Recognitions

Awards

Jul 2024	Winner in the TDL Challenge at ICML — Great Contributor	—
Oct 2019	Top 100 recognition @ <i>IEEE-Xtreme</i>	—

Grants

Jul 2025	Scholarship for EEML	300 EUR
Jul 2025	Scholarship for ELLIS Probabilistic ML school	250 GBP
Jun 2025	Scholarship for LOGML	250 EUR
Nov 2024	<i>Visit Grant</i> by DDSA (<i>Danish Data Science Academy</i>)	500 EUR
Jun 2024	<i>Travel Grant</i> by ELLIS Amsterdam for ICML	500 EUR

Honors

Aug 2025	Acceptance (w/ poster presentation) to ELLIS Doctoral Symposium	—
Jun 2025	Accepted to LOGML on project “ <i>Beyond VC Dimension</i> ”	—

Community Service

Conference & Workshop Organization

2026	Organization committee of LOGML 2026
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Teaching

Teaching Assistant

VU Amsterdam | Computer Science

2025	XM_40012: <i>Machine Learning for the Quantified Self</i>
2025	XB_0085: <i>Text Mining for AI</i>
2024	XM_0083: <i>Deep Learning</i> <i>Instructor:</i> Prof. Peter Bloem <i>Website/Course materials:</i> DL @ VU
2024	X_400111: <i>Evolutionary Computing</i>